MIL-P-60337 (MU) M
15 February 1966
(see 6.8)

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M47 AND M47B2
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling, and packing for percussion primer, M47 and M47B2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-P-223 - Powder, Black
MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607- Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defectives
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 1390

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

8839455 - Primer, Percussion M47 Assembly
8839460 - Primer, Percussion M47B2 Assembly
8858934 - Box, Packing, Ammunition, for Primers M47 or M47B2
8858940 - Box, Set-up, Packing Ammunition for Primers

PUBLICATIONS

ORDNANCE CORPS

IEL-8839455 - Index of Inspection Equipment Lists (M47)
IEL-8839460 - Index of Inspection Equipment Lists (M47B2)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications.

3.2 Primer assembly.-The primer assembly shall comply with all requirements specified on drawings (dwgs.) 8839455 and 8839460 as applicable and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of black powder at the loading station at the time of loading, shall not exceed 0.5 percent, when determined as specified in 4.3.3.

3.4 Head assembly

3.4.1 Non-functioning.-The head assembly shall not function by an applied energy of 47.52 inch ounces minimum (min.) when tested as specified in 4.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

3.4.2 Functioning.-The head assembly shall function by an applied energy of 159.20 inch ounces maximum (max.) when tested as specified in 4.3.1.2.

3.5 Primer assembly

3.5.1 Functioning.-The primer assembly shall function without audible hangfire, when tested as specified in 4.3.2.

3.5.2 Pressure resistance.-The primer assembly shall withstand a pressure of 40,000 plus or minus 4,000 pounds per square inch without evidence of rupture, or any part of the primer becoming detached, when tested as specified in 4.3.2.

3.6 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time of the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 Government verification.-Using the contractors written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of percussion primers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, interhcnagability, or storage as determined by the Government shall not necessitate changing the lot interfix number. The examinations and tests listed herein shall be performed on inspection lots submitted in accordance with Standards MIL-STD-105, MIL-STD-1235, or

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

MIL-STD-414 as applicable. The criteria and procedures for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168. Each lot shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. Primer, percussion M61 (herein after referred to as primer element) of one lot interfix number from one manufacturer.

4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used for defects other than critical, if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Head, prior to assembling battery cup and firing plug (see dwg. 8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Percussion element missing, cocked or inverted.....	Visual	01001
Minor:	AQL 1.0 percent		
201.	Seal between percussion element and head missing or incomplete...	Visual	01002
202.	Evidence of poor workmanship (see 3.6).....	Visual	01003

Source: https://assist.dla.mil -- Downloaded: 202610-23T00:29Z
Check the source to verify that this is the current version before use.

4.2.2.2 Head, prior to assembling to body (see dwg. 8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Firing plug or battery cup above flush with surface of head	Straight-Edge	02001
Major:	AQL 0.65 percent		
101.	Firing plug missing or inverted	Visual	02002
102.	Flash hole in head completely obstructed	Visual	02003
103.	Battery cup loose or missing	Manual	02004
Minor:	None defined		

4.2.2.3 Body, prior to assembly of liner and head assembly (see dwg. 8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Any flash hole missing from body	Gage	03001
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	03002

4.2.2.4 Body, prior to assembling head assembly (see dwg. 8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

Major:	AQL 0.40 percent		
101.	Weight of charge.....	Balance	04001
102.	Liner fails to cover all holes...	Visual	04002
Minor:	AQL 1.0 percent		
201.	Foreign matter in charge.....	Visual	04003
202.	Depth of diaphragm.....	Gage	04004

4.2.2.5 Assembly, prior to crimping (see dwg.8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 0.65 percent		
201.	Thread sealing compound missing..	Visual	05001

4.2.2.6 Assembly (see dwg. 8839455 or 8839460).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Head assembly loose on body.....	Manual	06001
*102.	Any flash hole not completely sealed.....	Visual	06002
Minor:	AQL 2.50 percent		
201.	Total length, max.....	Gage	06003
202.	Diameter of flange, max.....	Gage	06004
203.	Diameter of front end, max.....	Gage	06005
204.	Crimp of body to head missing or incomplete.....	Visual	06006
205.	Marking misleading or unidentifiable.....	Visual	06007
206.	Evidence of poor workmanship (see 3.6).....	Visual	06008

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

*Sketches 1, 2 and 3 on Figure 1 indicate acceptable primers. Furthermore, a primer with holes displaying the conditions 4, 5 or 6 may be classed acceptable provided that the visible liner is thoroughly impregnated with paint or lacquer and not more than 20 percent of the holes have this condition.

4.2.2.7 Set-up box, prior to sealing (see dwg. 8858940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Any packing material missing.....	Visual	07001
202.	Primer missing.....	Visual	07002
203.	Separator damaged to the extent that primers touch each other....	Visual	07003

4.2.2.8 Sealed set-up box (see dwg. 8858940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	None defined		
Minor:	AQL 1.5 percent		
201.	Sealing strip torn, incomplete badly wrinkled, or otherwise fails to seal carton.....	Visual	08001
202.	Carton torn, cut or punctured....	Visual	08002
203.	Marking missing, misleading or unidentifiable.....	Visual	08003
204.	Contents loose.....	Manual	08004

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.2.2.9 Sealed barrier bag (see dwg. 8858940).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bag punctured, torn or cut.....	Visual	09001
102.	Seal mislocated or incomplete....	Visual	09002
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable.....	Visual	09003

4.2.2.10 Packing box, prior to sealing (see dwg. 8858934).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Carton missing.....	Visual	10001
Minor:	None defined		

4.2.2.11 Packing box, sealed (see dwg. 8858934).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Box broken, board loose or protruding nail.....	Visual	11001
102.	Strapping missing, broken or loose.....	Visual	11002
103.	Cover improperly assembled.....	Visual	11003
Minor:	AQL 1.5 percent		
201.	Marking misleading or unidentifiable.....	Visual	11004
202.	Strapping improperly engaged or mislocated.....	Visual	11005
203.	Contents loose.....	Visual	11006

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

4.2.3 Testing

4.2.3.1 Head assembly

4.2.3.1.1 Non functioning (see 3.4.1) Critical defect - Code No. 02005.

4.2.3.1.1.1 Beginning with the first lot produced, and continuing until three consecutive lots have complied with the acceptance criteria specified, a sample of 150 head assemblies shall be randomly selected from each lot for the test. If any sample fails to comply with the requirement the lot shall be rejected. The test shall be performed as specified in 4.3.1.1.

4.2.3.1.1.2 After three consecutive lots have met the criteria of 4.2.3.1.1.1, a sample of 50 head assemblies shall be randomly selected from each lot for this test. If one or more primers fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.1.1 using equipment in accordance with 4.2.4. Primers subjected to this test shall be destroyed.

4.2.3.1.2 Functioning (see 3.4.2) Major defect - Code No. 02006.

4.2.3.1.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified two hundred and fifty samples shall be randomly selected from each lot for this test. If three or more primers fail to comply with the applicable requirements, the lot shall be rejected. If one or two defects are found, a second sample of five hundred primers shall be tested. If the combined number of defectives in both the first and second sample is three or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.3.1.2.2 After three consecutive lots have met the criteria of 4.2.3.1.2.1, a sample of 125 primers shall be randomly selected from each lot for this test. If two or more primers fail to comply with the applicable requirement, the lot shall be rejected. If only one defect is found, a second sample of 250 primers shall be tested. If the combined number of defectives in both the first and second sample is two or more, the lot shall be rejected. The test shall be performed as specified in 4.3.1.2 using equipment in accordance with 4.2.4.

4.2.3.2 Primer assembly functioning and pressure resistance (see 3.5) Major defect - Code No. 06009.-A sample of 25 primers shall be selected from each lot for this test. If any primer

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

fails to comply with the applicable requirement, the lot shall be rejected. The test shall be performed as specified in 4.3.2 using equipment that is in accordance with 4.2.4.

4.2.3.3 Check test for deterioration of primer element.-If the total elapsed time between original acceptance of the primer element lot and the assembly of that lot in to the primer exceeds two years, or if the primer elements have been subjected to adverse conditions for any period of time since previous test, the primer element lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable primer element specification immediately before the primer element lot is assembled into the primers (see 6.5).

4.2.3.4 Moisture content of black powder (see 3.3) Major defect - Code No. 04005.-A sample of black powder shall be randomly selected at the loading station from each eight hours production. The black powder shall be tested as specified in 4.3.3 and any failure to comply with the applicable requirement shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.5 Heat seal test for sealed bag (see dwg. 8858940) Major defect - Code No. 09004.- Sampling, acceptance criteria and test procedures for this test shall be in accordance with Specification MIL-P-116.

4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 8839455 or 8839460 as applicable identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment.-The contractor shall make available to the Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures

4.3.1 Head assembly

4.3.1.1 Non-functioning.-The loaded heads, prior to assembling to the body, shall be assembled into the test fixture (see 4.2.4). A steel ball weighing a min. of 7.92 ounces shall be dropped from a height to give the required impact force (approximately 6 inches).

4.3.1.2 Functioning.-The test shall be performed as specified in 4.3.1.1 except that the weight of the ball shall be 7.96 ounces maximum, and the height shall be adjusted to give the required impact force (approximately 20 inches).

4.3.2 Primer assembly functioning and pressure resistance.-The loaded primer assemblies shall be assembled into the test fixture (see 4.2.4). The fixture shall be activated and observation shall be made for firing and audible hangfire. Upon completion of firing the primer shall be removed and observation shall be made for ruptured and broken or detached parts. (see 6.6 and 6.7).

4.3.3 Determination of moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level C.-The primer assemblies shall be preserved and packaged in accordance with dwg. 8858940.

5.2 Packing

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

5.2.1 Level C.-The cartons shall be packed in accordance with dwg. 8858934.

5.3 Marking.-Marking shall be in accordance with dwg. 8858940 and 8858934.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists, (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 Inspection equipment lists.-Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF - Contractor's design responsibility on Army format in accordance with MIL-D-45608.
CDCF - Contractor's design responsibility on contractor format.
AD - Army design.
ADMU - Army design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

Source: https://assist.dla.mil -- Downloaded: 202640-23T00:29Z
Check the source to verify that this is the current version before use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs (see 6.3.6) or commercial equipment selections (see 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 Specification control drawing.-Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagramatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings must be described in sufficient detail to permit identification and evaluation by the Government (see 6.3.5). Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors normally are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.6 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.3.6 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL-D-45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black lines on white background and shall provide a legible print produced via the 35mm microfilm process. Contractors shall submit one copy of the final design as a rolled set or flat set. Designs shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Unless otherwise specified on the IEL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Check test for possible deterioration of percussion artillery primers.-The following tests are designed to prevent the loading of artillery primers which have deteriorated into a major ammunition item. The test shall be performed on normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the artillery primers are loaded does not require that the check test be performed, the contracting officer should insert a provision for the performance of the check tests in the contract.

6.4.1 Head assembly

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.4.1.1 Non-functioning (see 3.4.1).-The sampling plan for the test shall be in accordance with Table I. Any head assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.3.1.1, except that the flash tube and the head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable. All head assemblies so tested shall be scrapped.

TABLE I

Sample	Sample Size	Combined Samples Size	Acceptance Number	Rejection Number
First	20	20	*	2
Second	20	40	*	2
Third	20	60	0	2
Fourth	20	80	1	3
Fifth	20	100	1	3
Sixth	20	120	1	3
Seventh	20	140	2	3

*No acceptance on this sample

6.4.1.2 Functioning (see 3.4.2).-The sampling plan for the test shall be in accordance with Table II. Any head assembly which fails to comply with the applicable requirement shall be classed defective. The test shall be performed as specified in 4.3.1.2, except that the flash tube and and the head assembly shall be separated in a manner that will not damage the head assembly to the extent that the validity of the test results would be questionable.

TABLE II

Sample	Sample Size	Combined Samples Size	Acceptance Number	Rejection Number
First	75	75	*	2
Second	75	150	0	3
Third	75	225	0	3
Fourth	75	300	1	4
Fifth	75	375	2	5
Sixth	75	450	2	5
Seventh	75	525	2	5

*No acceptance on this sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

6.4.2 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the flash tube by any suitable means. The moisture content of the powder shall be determined in accordance with the method specified in Specification MIL-P-223. If the moisture content of any of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected. Since black powder is hygroscopic, it is advisable to place the powder in a dry, air-tight container immediately upon removal, and to run the determination as soon as possible thereafter.

6.5 Cost of check tests.-The contracting officer will arrange for the contractor to be reimbersed for the expense incurred in the performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the primer element or the contractor loading and assembling the primer, and shall not consitute a basis for rejection against either contractor, except where deteriortation has occurred as a direct result of carelessness in handling, storage, etc, while the primers were under the jurisdiction of either contractor.

6.6 Primer assembly functioning and pressure resistance.- In the event that a functioning test fixture is not available, ten sample primers shall be shipped to a Government proving ground and fired as part of a complete round for which it is standard at service pressure, from a weapon for which the round is standard. If any sample primer fails to comply with the applicable requirement, the lot shall be rejected.

6.7 Functioning test reports.-In addition to the normal distribution of records, one copy of primer functioning test results shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND2, Dover, New Jersey.

6.8 Supersession data.-This specification supersedes in part, Specification MIL-P-20466 (Ord) dated 4 December 1951, for Primers, Percussion, Manufacture, Loading, Assembling and Packing.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1390-A536

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

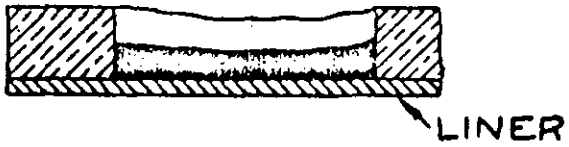
ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

NO. 1
PREFERRED

NO. 2
ACCEPTABLE

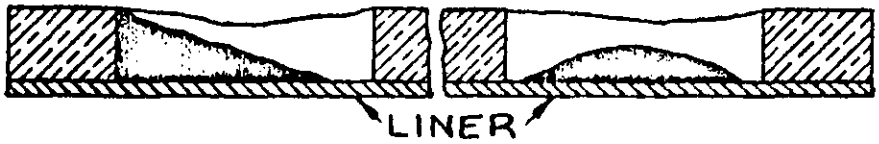


NO. 3
ACCEPTABLE



NO. 4
NOT ACCEPTABLE

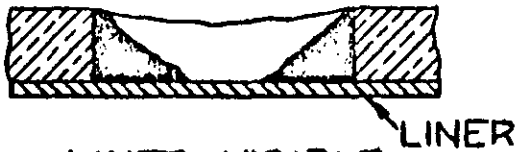
NO. 5
NOT ACCEPTABLE



LINER VISIBLE

LINER VISIBLE OR
NOT SEALED TO WALL

NO. 6
NOT ACCEPTABLE



LINER VISIBLE

FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.

ORDNANCE CORPS

ENGINEERING ORDER

ENGR ORDER NO. 41021-S

FOR REGULATION PERTAINING TO THE USE OF THIS FORM SEE ORDM-4-4 MANUAL

SHEET 1 OF 1 SHEETS

TO:

DATE: 15 February 1966 M

FROM: PICATINNY ARSENAL, DOVER, NEW JERSEY

COMMAND SYMBOL SMUPA-ND5

YOU ARE HEREBY NOTIFIED OF THE FOLLOWING ENGINEERING ACTION. TO BE APPLIED TO PROCUREMENT OR PRODUCTION ONLY UNDER INSTRUCTIONS ISSUED BY THE CONTRACTING AGENCY.

DESCRIPTIVE DATA

MATERIAL (ORD ITEM) Primer, Percussion, M47 & M47B2 Metal Parts For (MIL-P-60338 (MU)

AUTHORITY AMCR 10-22 22 December 1962

REASON FOR ENGINEERING ACTION:

Initial release of specification

ENGINEERING ACTION:

Specification MIL-P-60338 (MU) dated 15 February 1966 is hereby released for use in connection with procurement by the Army Munitions Command.

YES	NO		YES	NO	
	X	MODIFICATION TO PRIOR PRODUCTION FEASIBLE		X	DAMWO RECOMMENDED
	X	MODIFICATION RECOMMENDED/REQUIRED			

APPLICATION

☐ EXISTING CONTRACTS/ORDERS NOW IN PRODUCTION ☒ FUTURE CONTRACTS/ORDERS NOT YET IN EXISTENCE

☐ EXISTING CONTRACTS/ORDERS NOT YET IN PRODUCTION

CLASSIFICATION

URGENCY OF CHANGE

MANDATORY. CLASS III (CLASS I & II AFFECTS IMWO)

☐ OPTIONAL CLASS V ☐ INFORMATIONAL CLASS VI

APPLICATION

END ITEM REF. Various cartridges for 75MM Rifles

☒ INITIAL RELEASE

DISTRICT ACKNOWLEDGMENT

RECEIPT ACKNOWLEDGED BY:

DATE

SIG: CONTRACTING OFFICER OR AUTH. REP. ORD DIST.

AUTHORIZATION SECURED

INDICATE:

☐ OCO

☒ COMMAND

APPROVED BY:

M. P. FORSYTHE

CLASS I - PRIOR ITEM IS UNSUITABLE FOR ISSUE

CLASS II - PRIOR ITEM IS UNSUITABLE FOR COMBAT USE

CLASS III - NON INCORPORATION MAY CREATE DIFFICULTY

CLASS IV - INCORPORATION PROVIDES IMPROVEMENT

CLASS V - INCAPABLE OF AFFECTING PART

OD FORM 1963 FEB 66 1-66

☆U.S. GOVERNMENT PRINTING OFFICE:1972-714-164/13868

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:29Z
Check the source to verify that this is the current version before use.